- It is important that acid producing foods need to be balanced with alkali producing foods to reduce the risk of urine stone formation.

- When stones are composed of calcium phosphate or oxalate the urine is alkaline and hence acid ash diet is used to balance it.

Foods included: cereals, meats, fish, eggs, nuts, pulpy and pulses.

Foods avoided: milk, fruit and vegetables.

- As uric acid and cystine stones are formed by an acidic urinary environment the condition is treated with an alkali ash diet to neutralize the acidic urine.

Foods included: milk, fruit and vegetables.

Foods avoided: cereals, organ meats, fish, eggs, nuts, pulpy and pulses.

Dietary management

The diet is modified to reduce the intake of nutrients that lead to the formation of a particular type of stone.

1. Calcium stones

- A low calcium diet of approximately 400 mg/day is recommended for people who are not at risk for bone loss.

- The lower level mainly is achieved by restricting dairy products, which are the main dietary sources of calcium.

- If a stone is calcium phosphate, additional sources of phosphorous (e.g., meat, legumes and nuts) should be controlled.

- If a stone is calcium oxalate, foods high in oxalate like spinach, tomatoes, sweet potato, strawberry, chocolate, wheat products, nuts and tea should be avoided.

Other dietary factors

Other dietary factors to consider in the case of calcium stones are sodium fluid and fibre intake.

- High intake of salt increases the amount of calcium excretion in the urine, thus causing hypercalciuria.

- Diluting the urine by drinking plenty of fluids is beneficial in preventing all types of kidney stones.

- Foods high in phytates like whole wheat, soya help prevent the formation of calcium oxalate salts.

2. Uric acid stones

- A low purine diet is recommended to reduce uric acid.

- Foods to be avoided: organ meat, chicken, legumes, alcoholic beverages, spinach and cauliflower.

3. Cystine stones

- Cystine is derived from the essential amino acid methionine, so a low protein diet is preffered.